Statistics for Data Science - 2

Week 6 graded Assignment

Solution

1. The joint pdf of two continuous ranodm variables X and Y is given by

$$f_{XY}(x,y) = \begin{cases} 4xy & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Are X and Y independent?

1. Yes
2. No

Solution:
First we will calculate the marginal densities of X and Y.
For $0 \leq x \leq 1$

$$\begin{aligned} f_X(x) &= \int_0^1 f_{XY}(x,y)dy \\ &= \int_0^1 4xydy \\ &= 2xy^2\Big|_0^1 \\ &= 2x \end{aligned}$$

For $0 \leq y \leq 1$

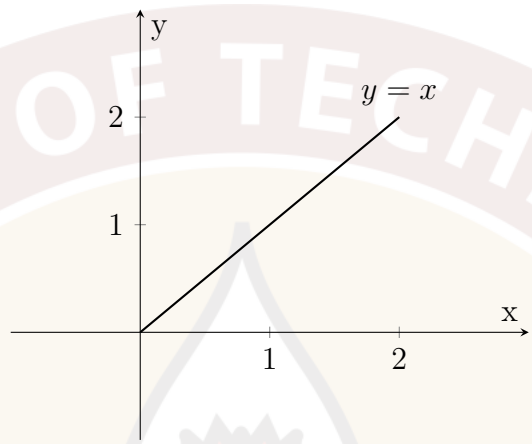
$$\begin{aligned} f_Y(y) &= \int_0^1 f_{XY}(x,y)dx \\ &= \int_0^1 4xydx \\ &= 2x^2y\Big|_0^1 \\ &= 2y \end{aligned}$$

Therefore,

$$f_X(x).f_Y(y) = 4xy = f_{XY}(x,y)$$

It implies that X and Y are independent random variables.

2. Let $(X, Y) \sim \text{Uniform}(D)$, where $D = \{(x, y) : (x-k)^2 + (y-k)^2 \leq r\}$. Calculate $P(X \geq Y)$.
Solution:



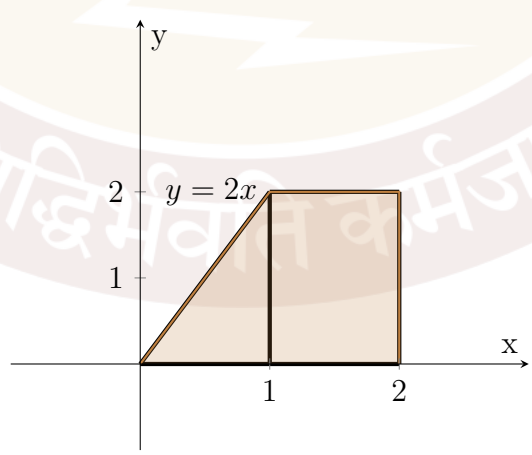
The region $X \geq Y$ will be the lower half part of the circle.

Therefore,

$$\begin{aligned} P(X \geq Y) &= \frac{\text{Area of lower half circle}}{\text{Area of the circle}} \\ &= \frac{\pi(1)^2/2}{\pi(1)^2} \\ &= \frac{1}{2} \end{aligned}$$

---

3. Let $(X, Y) \sim \text{Uniform}(D)$, where $D = \{(x, y) : y \leq 2x, 0 < x < 1, 0 < y < 2\} \cup [1, 2] \times [0, 2]$. Find the marginal density of X.



D denotes the area of the support(X, Y).
Area of $D = \frac{1}{2} \times 1 \times 2 + 1 \times 2 = 3$

Since $(X, Y) \sim \text{Uniform}(D)$, it implies that

$$f_{XY}(x, y) = \frac{1}{3}, \qquad x, y \in D$$

We know that $f_X(x) = \int f_{XY}(x, y)dy$

For $0 < x < 1$

$$\begin{aligned} f_X(x) &= \int_0^{2x} \frac{1}{3} dy \\ &= \frac{1}{3} y \Big|_0^{2x} \\ &= \frac{2x}{3} \end{aligned}$$

For $1 < x < 2$

$$\begin{aligned} f_X(x) &= \int_0^{2} \frac{1}{3} dy \\ &= \frac{1}{3} y \Big|_0^{2} \\ &= \frac{2}{3} \end{aligned}$$

Therefore, marginal density of X is given by

$$f_X(x) = \begin{cases} \frac{2x}{3} & 0 \leq x \leq 1 \\ \frac{2}{3} & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

---

4. The joint pdf of two random variables X and Y is given by

$$f_{XY}(x, y) = \begin{cases} 24xy & 0 \leq x \leq 1, 0 \leq y \leq 1, x + y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

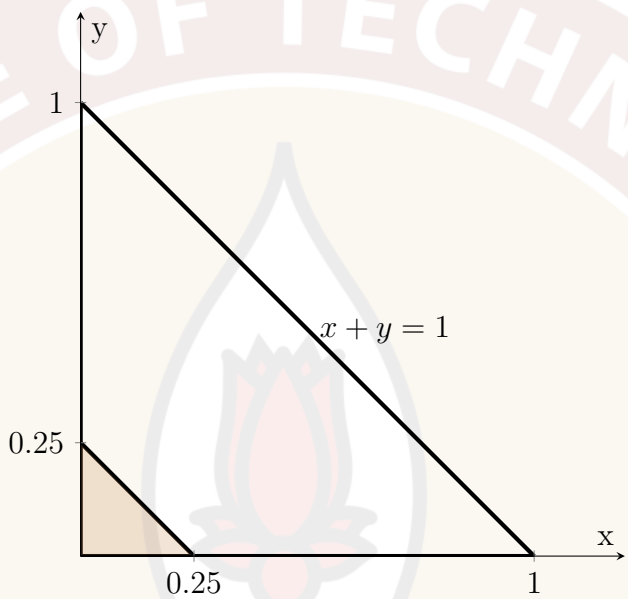
Choose the correct option(s).

(a) $P(X + Y \leq \frac{1}{4}) = \frac{1}{2}$
(b) $P(X + Y \leq \frac{1}{2}) = \frac{1}{16}$

(c) X and Y are independent random variables.

(d) X and Y are dependent random variables.

Solution:
Option (a)



Orange region will denote $X + Y \leq \frac{1}{4}$. Now,

$$P(X + Y \leq \frac{1}{4}) = \int_{y=0}^{1/4} \int_{x=0}^{1/4-y} f_{XY}(x, y) dx dy$$

$$= \int_{y=0}^{1/4} \int_{x=0}^{1/4-y} 24xy dx dy$$

$$= \int_{y=0}^{1/4} 12x^2 y \Big|_{x=0}^{1/4-y} dy$$

$$= \int_{y=0}^{1/4} 12y\left(\frac{1}{4} - y\right)^2 dy$$

$$= \int_{y=0}^{1/4} \frac{12}{16} y(1-4y)^2 dy$$

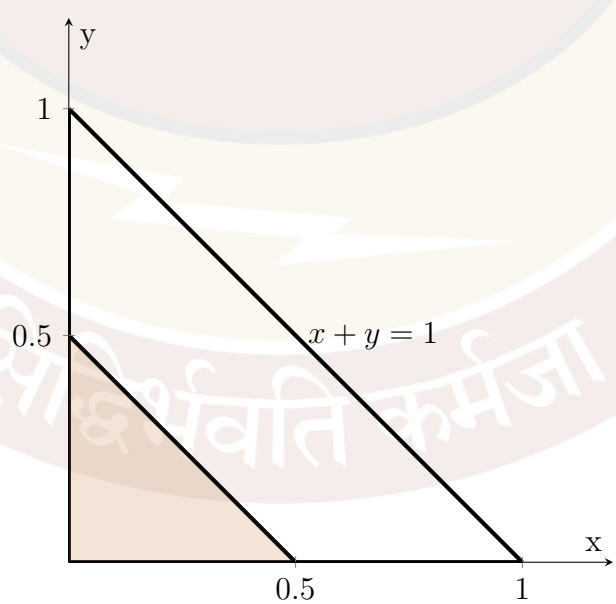
$$= \frac{3}{4}\int_{y=0}^{1/4} y(1+16y^2-8y)dy$$

$$= \frac{3}{4}\left(\frac{y^2}{2} + 4y^4 - \frac{8y^3}{3}\right)\Bigg|_{y=0}^{1/4}$$

$$= \frac{3}{4}\left(\frac{1}{32} + \frac{1}{64} - \frac{1}{24}\right)$$
$$= \frac{3}{4}.\frac{1}{192} = \frac{1}{256}$$

Hence, option (a) is wrong.

<u>Option (b)</u>



Orange region will denote $X + Y \leq \frac{1}{2}$. Now,

$$P(X+Y \leq \frac{1}{2}) = \int_{y=0}^{1/2} \int_{x=0}^{1/2-y} f_{XY}(x,y)dxdy$$

$$= \int_{y=0}^{1/2} \int_{x=0}^{1/2-y} 24xydxdy$$

$$= \int_{y=0}^{1/2} 12x^2y \Big|_{x=0}^{1/2-y} dy$$

$$= \int_{y=0}^{1/2} 12y \left(\frac{1}{2} - y\right)^2 dy$$

$$= \int_{y=0}^{1/2} \frac{12}{4} y(1-2y)^2 dy$$

$$= 3 \int_{y=0}^{1/2} y(1+4y^2-4y)dy$$

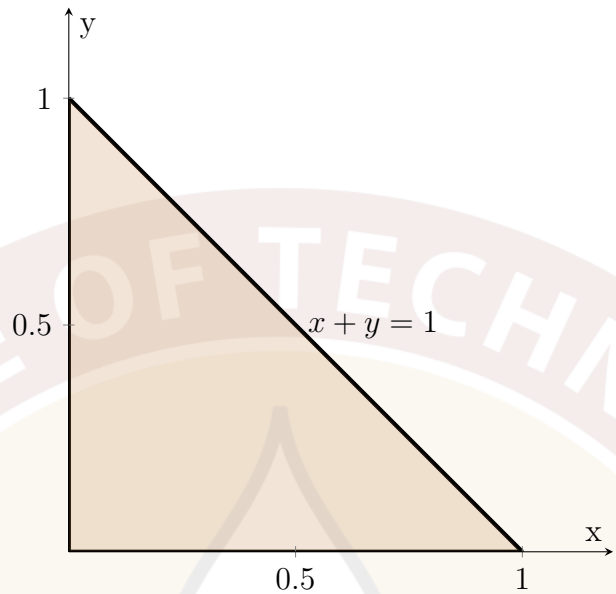
$$= 3 \left(\frac{y^2}{2} + y^4 - \frac{4y^3}{3}\right)\Big|_{y=0}^{1/2}$$

$$= 3 \left(\frac{1}{8} + \frac{1}{16} - \frac{1}{6}\right)$$

$$= 3 \times \frac{2}{96} = \frac{1}{16}$$

Hence, option (b) is correct.

<u>Option (c) and (d)</u>



For $0 < x < 1$

$$\begin{aligned} f_X(x) &= \int_{y=0}^{1-x} f_{XY}(x,y)dy \\ &= \int_{y=0}^{1-x} 24xydy \\ &= 12xy^2\Big|_{y=0}^{1-x} \\ &= 12x(1-x)^2 \end{aligned}$$

For $0 < y < 1$

$$\begin{aligned} f_Y(y) &= \int_{x=0}^{1-y} f_{XY}(x,y)dx \\ &= \int_{0}^{1-y} 24xydx \\ &= 12x^2y\Big|_{x=0}^{1-y} \\ &= 12y(1-y)^2 \end{aligned}$$

Therefore, $f_X(x).f_Y(y) = 144xy(1-x)^2(1-y)^2 \neq f_{XY}(x,y)$

Hence, X and Y are not independent.

5. The joint pdf of two random variables X and Y is given by

$$f_{XY}(x,y) = \begin{cases} 3xy(1-x) & 0 \le x \le 1, 0 \le y \le 2 \\ 0 & \text{otherwise} \end{cases}$$

Calculate $P(X > \frac{1}{2}|Y = 1)$.
Solution:
We know that

$$P(a < X < b|Y = y) = \frac{f_{XY}(a < X < b, y)}{f_Y(y)}$$

Now,

$$\begin{aligned} f_Y(y) &= \int_0^1 3xy(1-x)dx \\ &= \int_0^1 (3xy - 3x^2y)dx \\ &= \left(\frac{3x^2y}{2} - x^3y\right)\Bigg|_0^1 \\ &= \frac{3y}{2} - y = \frac{y}{2} \end{aligned}$$

Therefore, $f_Y(1) = \frac{1}{2}$
Now,

$$\begin{aligned} P(X > \frac{1}{2}|Y = 1) &= \frac{f_{XY}(X > \frac{1}{2}, Y = 1)}{f_Y(1)} \\ &= 2f_{XY}(X > \frac{1}{2}, Y = 1) \\ &= \int_{x=\frac{1}{2}}^1 2(3x(1-x))dx \\ &= 6\int_{\frac{1}{2}}^1 (x - x^2)dx \\ &= 6\left(\frac{x^2}{2} - \frac{x^3}{3}\right)\Bigg|_{\frac{1}{2}}^1 \\ &= 6\left(\frac{1}{2} - \frac{1}{3}\right) - 6\left(\frac{1}{8} - \frac{1}{24}\right) = 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

---

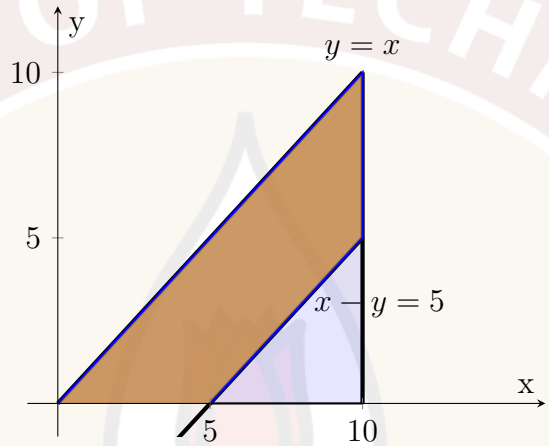
6. The amount of milk (in litres) in a shop at the beginning of any day is a random amount X from which a random amount Y (in litres) is sold during that day. Assume that the

joint density function of X and Y is given by

$$f_{XY}(x,y) = \begin{cases} \frac{1}{50} & 0 \le x \le 10, 0 \le y \le x \\ 0 & \text{otherwise} \end{cases}$$

Find the probability that amount of milk left at the end of day is less than 5 litres. Write your answer correct to two decimal points.

Solution:



X denotes the amount of milk at the beginning of any day and Y denotes the amount of milk which is sold during that day.
Therefore, amount of milk left at the end of the day will be denoted by $X - Y$.

To find: $P(X - Y < 5)$

In the diagram above, brown region denotes $X - Y < 5$ and brown + blue region denotes the support of X and Y.

Area of the support$(X, Y) = \frac{1}{2} \times 10 \times 10 = 50$.

Area of brown region = Area of support$(X, Y)-$ area of blue region

$\Rightarrow$ area of brown region $= 50 - \frac{1}{2} \times 5 \times 5 = \frac{75}{2}$

Therefore,

$$\begin{aligned} P(X - Y < 5) &= \frac{\text{area of brown region}}{\text{area of support}} \\ &= \frac{75/2}{50} \\ &= \frac{75}{100} \end{aligned}$$

7. The joint pdf of two continuous random variables X and Y is given by

$$f_{XY}(x,y) = \begin{cases} ke^{-(x+y)} & x \geq 0, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of $P(X \geq 5, Y \leq 5)$.

(a) e^{-10}

(b) $(e^{-5} - 1)e^{-5}$

(c) $(1 - e^{-5})e^{-5}$

(d) $(e^{-5} + 1)e^{-5}$

Solution:
We know that

$$\int\int_{\text{Supp}(X,Y)} f_{XY} dxdy = 1$$

Therefore,

$$\begin{aligned}
&\int_{y=0}^{\infty}\int_{x=0}^{\infty}(ke^{-(x+y)})dxdy = 1 \\
\Rightarrow &k\int_{y=0}^{\infty}\int_{x=0}^{\infty} e^{-y}e^{-x}dxdy = 1 \\
\Rightarrow &k\int_{y=0}^{\infty} e^{-y}(-e^{-x})\Big|_0^{\infty} dy = 1 \\
\Rightarrow &k\int_{y=0}^{\infty} e^{-y}(0+1)dy = 1 \\
\Rightarrow &k\int_{y=0}^{\infty} e^{-y}dy = 1 \\
\Rightarrow &k(-e^{-y})\Big|_0^{\infty} = 1 \\
\Rightarrow &k(0+1) = 1 \\
\Rightarrow &k = 1
\end{aligned}$$

To find: $P(X \geq 5, Y \leq 5)$

Now,

$$
\begin{aligned}
P(X \geq 5, Y \leq 5) &= \int_{y=0}^{5} \int_{x=5}^{\infty} (e^{-(x+y)}) dx dy \\
&= \int_{y=0}^{5} \int_{x=5}^{\infty} e^{-y} e^{-x} dx dy \\
&= \int_{y=0}^{5} e^{-y} (-e^{-x}) \Big|_{5}^{\infty} dy \\
&= \int_{y=0}^{5} e^{-y} (0 + e^{-5}) dy \\
&= (e^{-5}) \int_{y=0}^{5} e^{-y} dy \\
&= (e^{-5})(-e^{-y}) \Big|_{0}^{5} \\
&= (e^{-5})(-e^{-5} + 1) \\
&= (e^{-5})(1 - e^{-5})
\end{aligned}
$$

---

8. The joint pdf of two random variables X and Y is given by

$$
f_{XY}(x, y) = \begin{cases} \frac{1}{8}(x + y) & 0 \leq x \leq 2, 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}
$$

Find the value of $P\left(\frac{1}{2} \leq y \leq 1 \mid (X = \frac{1}{2})\right)$. Write your answer correct to two decimal points.

Solution:

We know that

$$
P(a < Y < b | X = x) = \frac{f_{XY}(X = x, a < Y < b)}{f_X(x)}
$$

Now,

$$
\begin{aligned}
f_X(x) &= \int_0^2 \frac{1}{8}(x + y) dy \\
&= \frac{1}{8}\left(xy + \frac{y^2}{2}\right)\Big|_0^2 \\
&= \frac{2x + 2}{8} = \frac{x + 1}{4}
\end{aligned}
$$

Therefore, $f_X(\frac{1}{2}) = \frac{3}{8}$

Now,

$$P(\frac{1}{2} \leq Y \leq 1 | X = \frac{1}{2}) = \frac{f_{XY}(X = \frac{1}{2}, \frac{1}{2} \leq Y \leq 1)}{f_X(\frac{1}{2})}$$

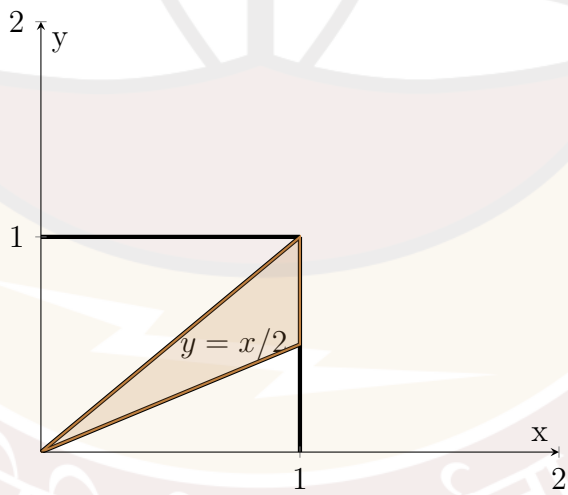
$$= \int_{1/2}^{1} \frac{8}{3} \left[\frac{1}{8} \left(\frac{1}{2} + y \right) \right] dy$$

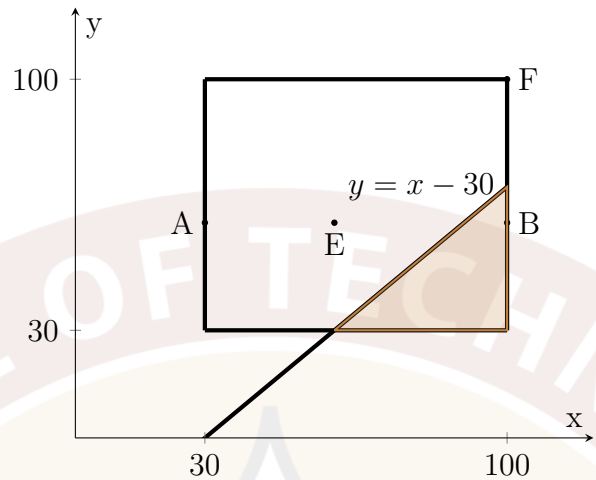
$$= \int_{1/2}^{1} \frac{1}{3} \left(\frac{1}{2} + y \right) dy$$

$$= \left(\frac{y}{6} + \frac{y^2}{6} \right) \Bigg|_{1/2}^{1}$$

$$= \left(\frac{1}{6} + \frac{1}{6} \right) - \left(\frac{1}{12} + \frac{1}{24} \right)$$

$$= \frac{1}{3} - \frac{1}{8} = \frac{5}{24} = 0.20$$





Use the following information to answer questions (9) and (10):

The joint density function of two random variables X and Y is given by

$$f_{XY}(x,y) = \begin{cases} \frac{2}{3}(x+1), & \text{for } 0 < x < 1, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

9. Are X and Y independent?

   (a) Yes

   (b) No

<u>Solution:</u>

First we will calculate the marginal densities of X and Y.
For $0 < x < 1$

$$\begin{aligned} f_X(x) &= \int_0^1 f_{XY}(x,y)dy \\ &= \int_0^1 \frac{2}{3}(x+1)\ dy \\ &= \frac{2}{3}(x+1)y\Big|_{y=0}^{1} \\ &= \frac{2}{3}(x+1) \end{aligned}$$

For $0 < y < 1$

$$\begin{aligned} f_Y(y) &= \int_0^1 f_{XY}(x,y)dx \\ &= \int_0^1 \frac{2}{3}(x+1)\ dx \\ &= \frac{2}{3}\left(\frac{x^2}{2}+x\right)\Bigg|_0^1 \\ &= \frac{2}{3}\times\frac{3}{2} = 1 \end{aligned}$$

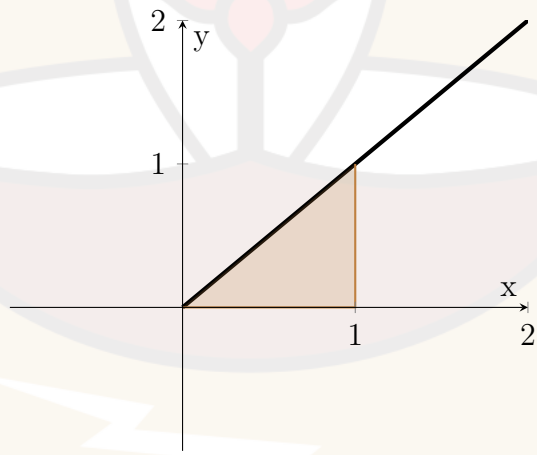
Therefore,

$$f_X(x).f_Y(y) = \frac{2}{3}(x+1) = f_{XY}(x,y)$$

It implies that X and Y are independent random variables.

10. Find the value of $P(X > Y)$. Write your answer correct to two decimal places.

<u>Solution:</u>



The shaded region here represents $X > Y$.

$$
\begin{aligned}
P(X > Y) = & \int_0^1 \int_y^1 \frac{2}{3}(x+1)\ dxdy \\
= & \frac{2}{3} \int_0^1 \left(\frac{x^2}{2} + x \right) \Bigg|_y^1 dy \\
= & \frac{2}{3} \int_0^1 \left(\frac{3}{2} - \frac{y^2}{2} - y \right) dy \\
= & \frac{2}{3} \left(\frac{3}{2}y - \frac{y^3}{6} - \frac{y^2}{2} \right) \Bigg|_0^1 \\
= & \frac{2}{3} \times \frac{5}{6} = 0.55
\end{aligned}
$$

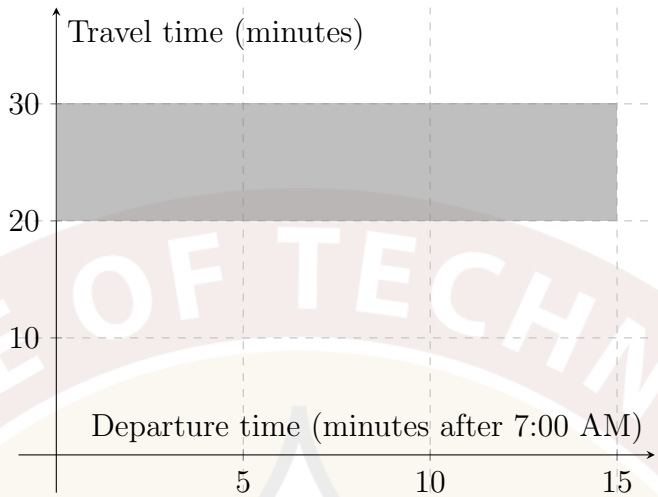
11. Riya leaves for school between 7:00 AM and 7:15 AM. She takes between 20 and 30 minutes to reach the school. Let X denote her time of departure, and let Y denote her travel time. Assuming that X and Y are continuous, independent and uniformly distributed, find the probability that she reaches the school before 7:30 AM. Enter the answer correct to two decimal places.

<u>Solution:</u>
Let X be the time of departure in minutes after 7:00 AM and is uniformly distributed between 0 and 15 minutes. So, $X \sim$Uniform$(0, 15)$.

Let Y be the travel time (in minutes), and is uniformly distributed between 20 and 30 minutes. So, $Y \sim$ Uniform$(20, 30)$

Then, density function of X and Y are:
$f_X(x) = \frac{1}{15}$ for $0 \leq x \leq 15$
$f_Y(y) = \frac{1}{10}$ for $20 \leq x \leq 30$

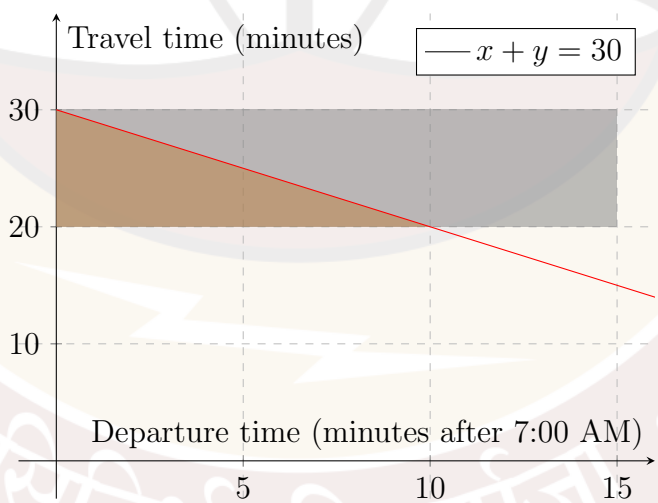
Since, X and Y are independent, their joint density is:
$f_{XY}(xy) = \frac{1}{15} \cdot \frac{1}{10} = \frac{1}{150}$



$f_{XY}(x, y)$

We need to find probability that she reaches the school before 7:30 AM i.e.

$$P(X + Y < 30)$$



Graph of $P(X + Y < 30)$

$$
\begin{aligned}
P(X+Y<30) &= \frac{\text{Area of required region}}{\text{Total area}} \\
&= \frac{\frac{1}{2}\cdot 10\cdot 10}{15\cdot 10} \\
&= \frac{50}{150} \\
&= \frac{1}{3} \\
&= 0.33
\end{aligned}
$$

Use the following information to answer questions (12) and (13):

The joint density function of two random variables X and Y is given by

$$
f(x,y) = \begin{cases} k(x+2y) & \text{for } 0<x<1, 1<y<2 \\ 0 & \text{otherwise} \end{cases}
$$

12. Find the value of k. Enter the answer correct to three decimal places.

<u>Solution:</u>
We know that

$$
\int\int_{\text{Supp}(X,Y)} f_{XY}\,dxdy = 1
$$

Therefore,

$$\int_1^2 \int_0^1 k(x+2y)\,dx\,dy = 1$$
$$\Rightarrow k \int_1^2 \frac{x^2}{2}\bigg|_0^1 + 2y\,dy = 1$$
$$\Rightarrow k\left\{\frac{1}{2} + 2\cdot\frac{y^2}{2}\bigg|_1^2\right\} = 1$$
$$\Rightarrow k\left\{\frac{1}{2} + 3\right\} = 1$$
$$\Rightarrow k\left\{\frac{7}{2}\right\} = 1$$
$$\Rightarrow k = \frac{2}{7}$$
$$\Rightarrow k = 0.285$$

13. Find the expectation of $(X|Y = 1.5)$ i.e., $E[X|Y = 1.5]$. Enter the answer correct to two decimal places.

<u>Solution:</u>
Firstly, find joint PDF of $(X|Y = 1.5)$.

$$P(X = x|Y = 1.5) = \frac{P(X = x, Y = 1.5)}{P(Y = 1.5)}$$
$$= \frac{\frac{2}{7}(x + 2\times 1.5)}{P(Y = 1.5)}$$
$$= \frac{\frac{2}{7}x + \frac{6}{7}}{P(Y = 1.5)} - - - - - - - eq(1)$$

Now, find the marginal of Y

$$
\begin{aligned}
P(Y=y) &= \int_0^1 f_{XY}(x,y), dx \\
&= \int_0^1 \frac{2}{7}(x+2y)\, dx \\
&= \int_0^1 \frac{2}{7}x + \frac{4}{7}y\, dx \\
&= \frac{2}{7}\cdot\frac{x^2}{2}\bigg|_0^1 + \left(\frac{4}{7}y\right)x\bigg|_0^1 \\
&= \frac{1}{7} + \frac{4}{7}y \\
&= \frac{1+4y}{7}
\end{aligned}
$$

$$
\Rightarrow P(Y=1.5) = \frac{1+4(1.5)}{7} = 1
$$

On putting $P(Y=1.5)=1$ in $eq(1)$

$$
P(X=x|Y=1.5) = \frac{2}{7}x + \frac{6}{7}
$$

Now,

$$
\begin{aligned}
E[X|Y=1.5] &= \int_0^1 x\left(\frac{2}{7}x + \frac{6}{7}\right)\, dx \\
&= \int_0^1 \frac{2}{7}\cdot x^2 + \frac{6}{7}\cdot x\, dx \\
&= \frac{2}{7}\cdot\frac{x^3}{3}\bigg|_0^1 + \frac{6}{7}\cdot\frac{x^2}{2}\bigg|_0^1 \\
&= \frac{2}{7}\cdot\frac{1}{3} + \frac{6}{7}\cdot\frac{1}{2} \\
&= \frac{2}{21} + \frac{6}{14} \\
&= \frac{22}{42} \\
&= \frac{11}{21} \\
&= 0.52
\end{aligned}
$$

14. Let X and Y have the following joint density function:

$$f_{XY}(x,y) = \begin{cases} a(x+y-3xy^2); & 0 \le x \le 1, 0 \le y \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the value of a such that $f_{XY}(x,y)$ is a valid PDF.
Answer : 2
Solution:
As we know that the integration of a pdf over the entire range is equal to 1,therefore,

$$\int_0^1 \int_0^1 a(x+y-3xy^2)dydx = 1$$

$$\int_0^1 a\left[xy + \frac{y^2}{2} - \frac{3xy^3}{3}\right]_0^1 dx = 1$$

$$\int_0^1 a\left(x + \frac{1}{2} - x\right) dx = 1$$

$$\int_0^1 \frac{a}{2} dx = 1$$

$$\left[\frac{ax}{2}\right]_0^1 = 1$$

$$\frac{a}{2} = 1$$

$$a = 2$$

(b) Calculate the value of $P\left(Y < \frac{X}{2}\right)$. Enter the answer upto two decimal places.
Answer : 0.37 , Range : 0.34 - 0.40
Solution :

$$P\left(Y < \frac{X}{2}\right) = \int_0^1 \int_0^{\frac{x}{2}} 2(x+y-3xy^2)dydx$$

$$P\left(Y < \frac{X}{2}\right) = \int_0^1 2\left[xy + \frac{y^2}{2} - \frac{3xy^3}{3}\right]_0^{\frac{x}{2}} dx$$

$$P\left(Y < \frac{X}{2}\right) = \int_0^1 2\left[\frac{x^2}{2} + \frac{x^2}{8} - \frac{x^4}{16}\right] dx$$

$$P\left(Y < \frac{X}{2}\right) = 2\left[\frac{x^3}{6} + \frac{x^3}{24} - \frac{x^5}{40}\right]_0^1$$

$$P\left(Y < \frac{X}{2}\right) = 2\left[\frac{1}{6} + \frac{1}{24} - \frac{1}{40}\right]$$

$$P\left(Y < \frac{X}{2}\right) = 2\left[\frac{20+5-3}{120}\right]$$

$$P\left(Y < \frac{X}{2}\right) = \frac{2 \times 22}{120}$$

$$P\left(Y < \frac{X}{2}\right) = 0.37$$

15. Let X and Y be two continuous random variables with joint PDF(probability density function) :

$$f_{XY}(x,y) = \begin{cases} xe^{-x(y+1)}; & x > 0, y > 0 \\ 0, & \text{otherwise.} \end{cases}$$

(i.) Find the marginal distribution of X.

(a) $f_X(x) = \begin{cases} xe^{(2x)}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$

(b) $f_X(x) = \begin{cases} xe^{(-2x)}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$

(c) $f_X(x) = \begin{cases} 1, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$

(d) $f_X(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$

Answer : (d)

Solution :

Integrating the function with respect to y, we get:

$$f_X(x) = \int_0^\infty xe^{-x(y+1)}dy$$

$$f_X(x) = \int_0^\infty x\left[e^{-xy}e^{-x}\right]dy$$

$$f_X(x) = xe^{-x}\int_0^\infty e^{-xy}dy$$

$$f_X(x) = xe^{-x}\left[\frac{e^{-xy}}{-x}\right]_0^{\infty}$$

$$f_X(x) = e^{-x}$$

$$\text{therefore, } f_X(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & \text{otherwise.} \end{cases}$$

(ii.) Find the value of $P(Y > 2 \mid X > 1)$.

(a) e^{-2}

(b) $\frac{1}{3}e^{-2}$

(c) $\frac{1}{3}e^{-3}$

(d) e^{-1}

Answer : (b)

Solution:

$$P(Y > 2|X > 1) = \frac{P(Y > 2, X > 1)}{P(X > 1)}$$

$$P(Y > 2|X > 1) = \frac{\int_1^{\infty}\int_2^{\infty} xe^{-x(y+1)}dydx}{\int_1^{\infty} e^{-x}dx}$$

$$P(Y > 2|X > 1) = \frac{\int_1^{\infty} xe^{-x}\left[\frac{e^{-xy}}{-x}\right]_2^{\infty} dx}{\left[\frac{e^{-x}}{-1}\right]_1^{\infty}}$$

$$P(Y > 2|X > 1) = \frac{\int_1^{\infty} e^{-x}\left[e^{-2x}\right] dx}{e^{-1}}$$

$$P(Y > 2|X > 1) = \frac{\int_1^{\infty} e^{-3x}dx}{e^{-1}}$$

$$P(Y > 2|X > 1) = \frac{\left[\frac{e^{-3x}}{-3}\right]_1^{\infty}}{e^{-1}}$$

$$P(Y > 2|X > 1) = \frac{\left[\frac{e^{-3}}{-3}\right]}{e^{-1}}$$

$$P(Y > 2|X > 1) = \frac{e^{-2}}{3}$$